COMPUTER VISION

Image Segmentation using Full Convolutional Neural Networks

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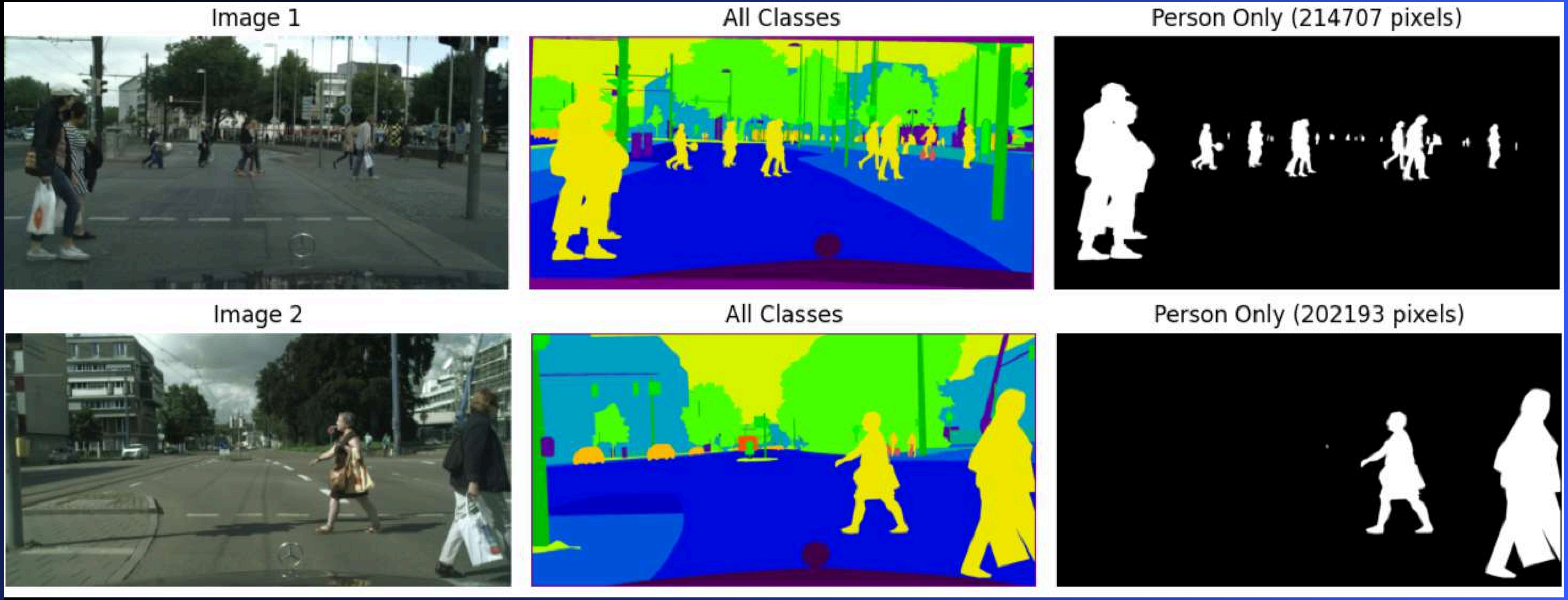
OUR DATASET

Cityscape

- street view images
- different coloured masks for objects
- detecting people based on ID (24)



OUR MODEL

Preparation:

- Matching pictures with masks
- Data augmentation
- Limit dataset (300 images)

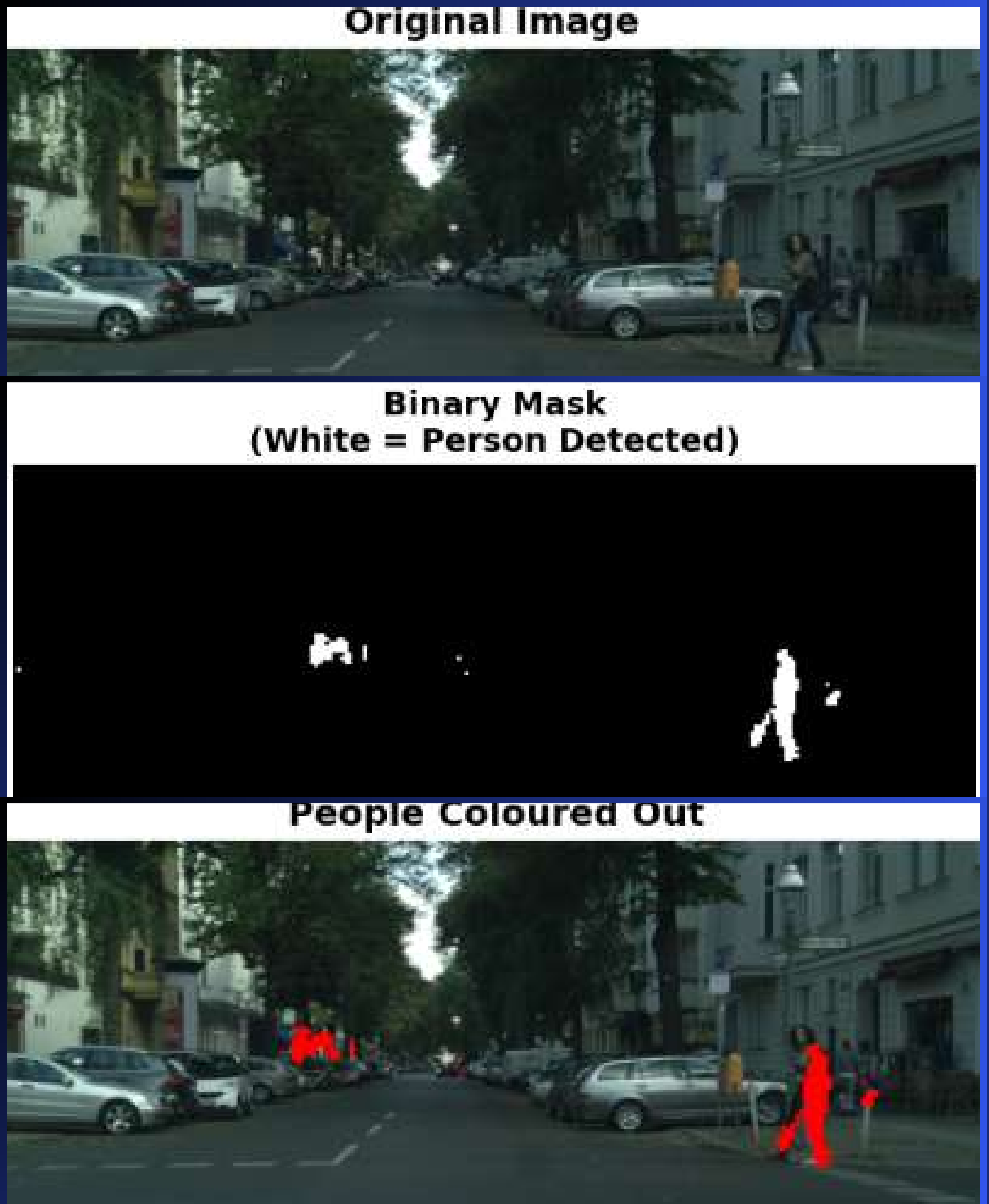
34 functional layers

- 26 distinct trainable layers (Conv2D and BatchNormalization)
- 8 non-trainable layers (MaxPool2D, Dropout, UpSampling2D, Input, and Concatenate)

U-Net Model:

- Pixel Level Precision
- "U-Shape" → WHAT is in the image and WHERE is it located
- Skip Connections → links encoder to decoder

batch_normalizatio… (BatchNormalizatio…	(None, 128, 2…, 32)	128	conv2d_38[0][0]
conv2d_39 (Conv2D)	(None, 128, 256, 32)	9,248	batch_normalizat…
batch_normalizatio… (BatchNormalizatio…	(None, 128, 256, 32)	128	conv2d_39[0][0]
max_pooling2d_8 (MaxPooling2D)	(None, 64, 128, 32)	0	batch_normalizat…
conv2d_40 (Conv2D)	(None, 64, 128, 64)	18,496	max_pooling2d_8[…
batch_normalizatio… (BatchNormalizatio…	(None, 64, 128, 64)	256	conv2d_40[0][0]
conv2d_41 (Conv2D)	(None, 64, 128, 64)	36,928	batch_normalizat…
batch_normalizatio… (BatchNormalizatio…	(None, 64, 128, 64)	256	conv2d_41[0][0]
max_pooling2d_9 (MaxPooling2D)	(None, 32, 64, 64)	0	batch_normalizat…
conv2d_4…			



OUR RESULTS

Issue:

- Humans are very small percentage of pictures (<1%)

→ predicting all black (no human) already has high accuracy

- Not all pictures have humans!

U-Net Model:

- High accuracy but best IoU only 0.1884 (difficult task)
- Can "see" people that are not even noticable for the human eye

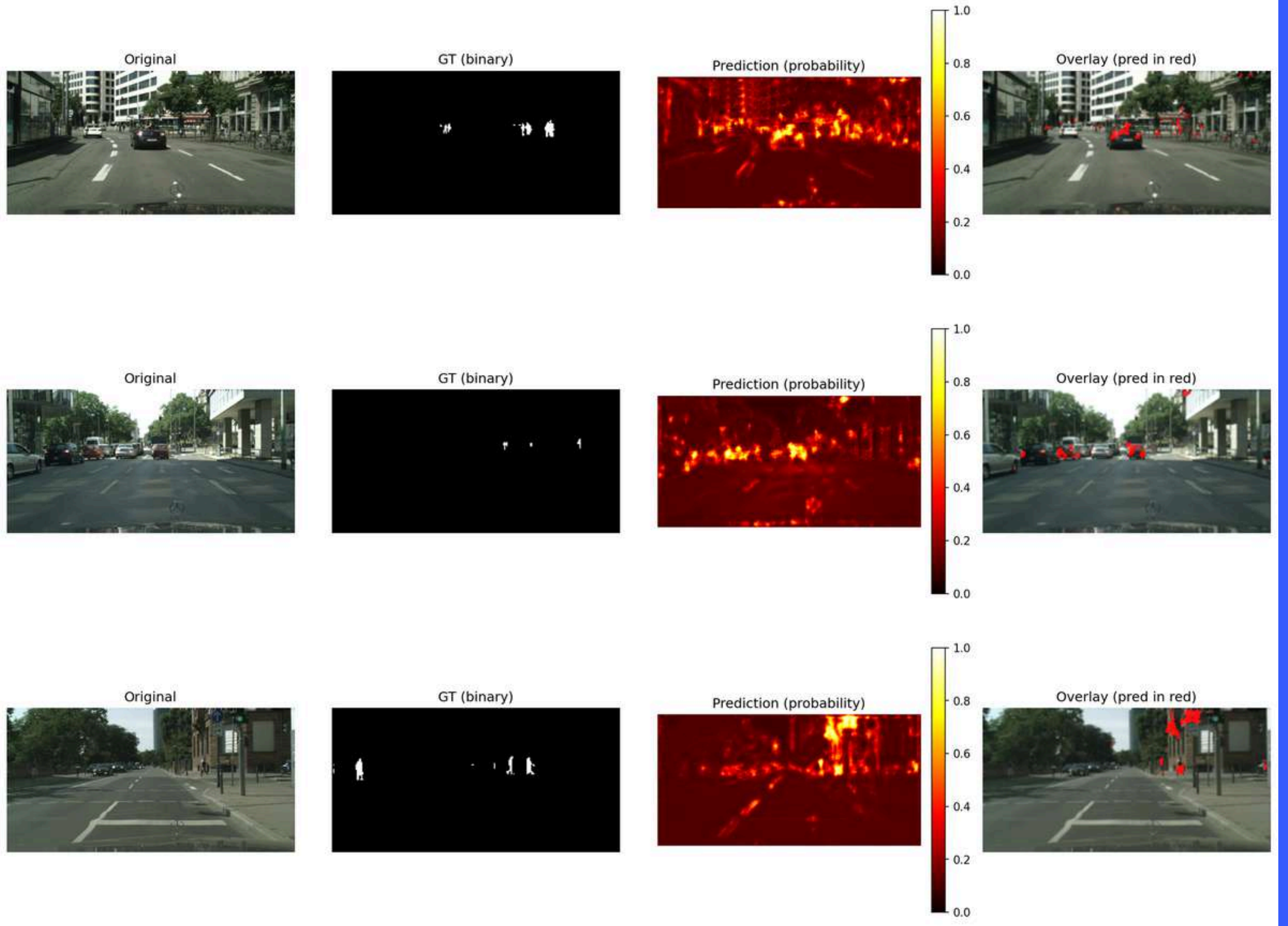
BASELINE MODEL

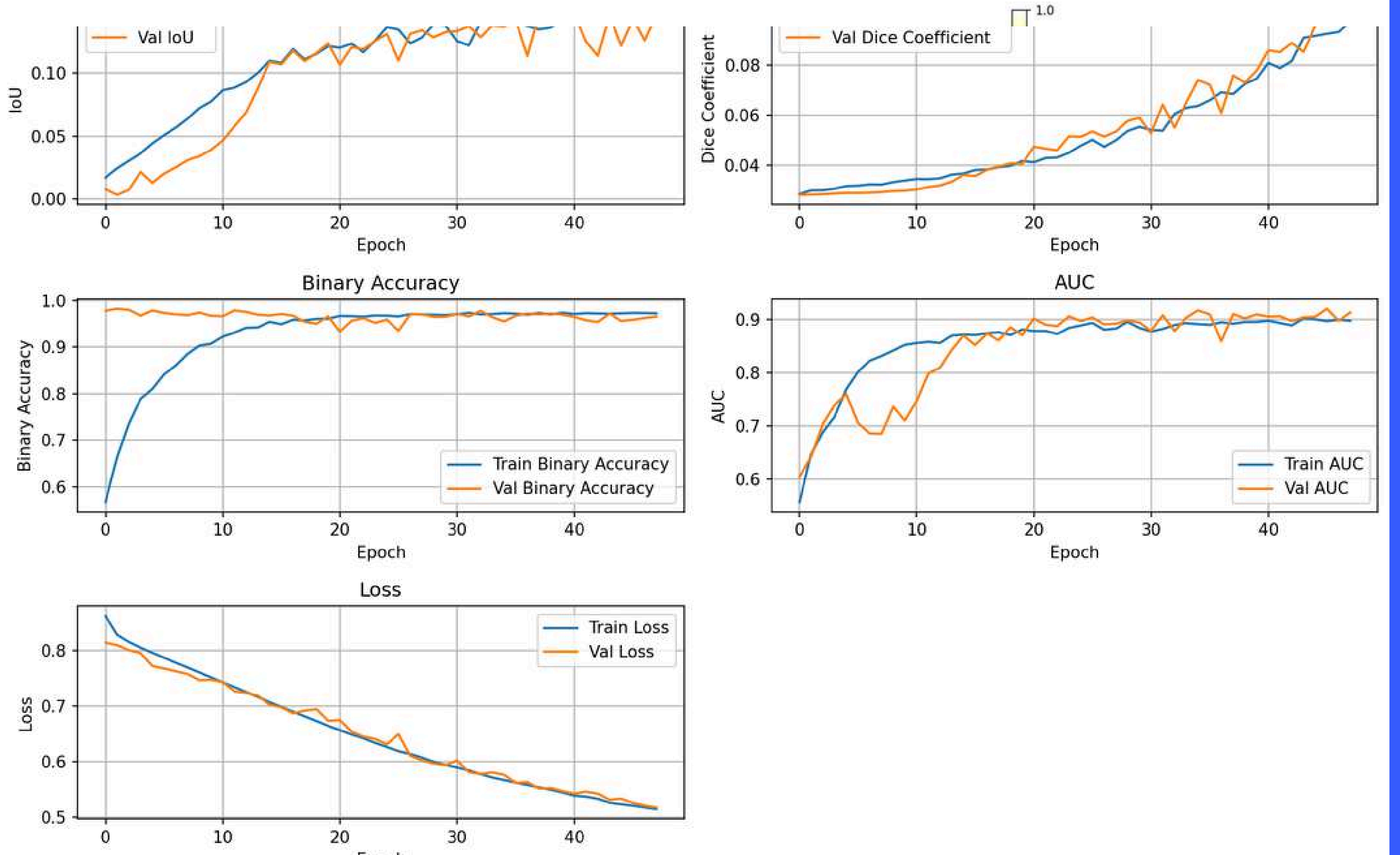
1

A lightweight FCN architecture was implemented as the baseline model, consisting of four convolutional layers followed by a 1×1 convolutional kernel for pixel-wise classification. The model contains 75,585 trainable parameters, making it compact enough for rapid experimentation but offering limited spatial feature extraction and context modeling for reliable person-mask segmentation.

2

Segmentation was performed specifically on the person class (label ID 24) using a binary setup (person vs. background). Accuracy, AUC, loss curves, and IoU were continuously monitored during training to assess model behaviour. This simplified FCN serves purely as a baseline reference and highlights the performance gap when compared to more advanced segmentation architectures used in subsequent experiments.





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FINAL MODEL COMPARISON (Best Validation Scores)
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| Model    |   Best Val IoU |   Best Val Dice |   Best Val AUC | Total Params    |   Inference Time (ms) |
|:--------|---------------:|----------------:|---------------:|:---------------|----------------------:|
| U-Net    |         0.1884 |          0.0547 |         0.9504 | 972,097         |                 37.15 |
| FCN      |         0.1612 |          0.0748 |         0.9004 | 75,585          |                 27.02 |
| FCN 1x1 |         0      |          0.0248 |         0.6492 | 4               |                  5.69 |

Analysis Metrics:
  - Best Val IoU (Intersection over Union): Primary metric for segmentation accuracy.
  - Best Val Dice (Dice Coefficient): Also known as F1-Score, closely related to IoU.
  - Best Val AUC (Area Under Curve): Measures the model's ability to discriminate between classes (person vs. background).
  - Total Params: A measure of model complexity (proxy for memory/disk size).
  - Inference Time: Measure of speed (critical for real-time applications).
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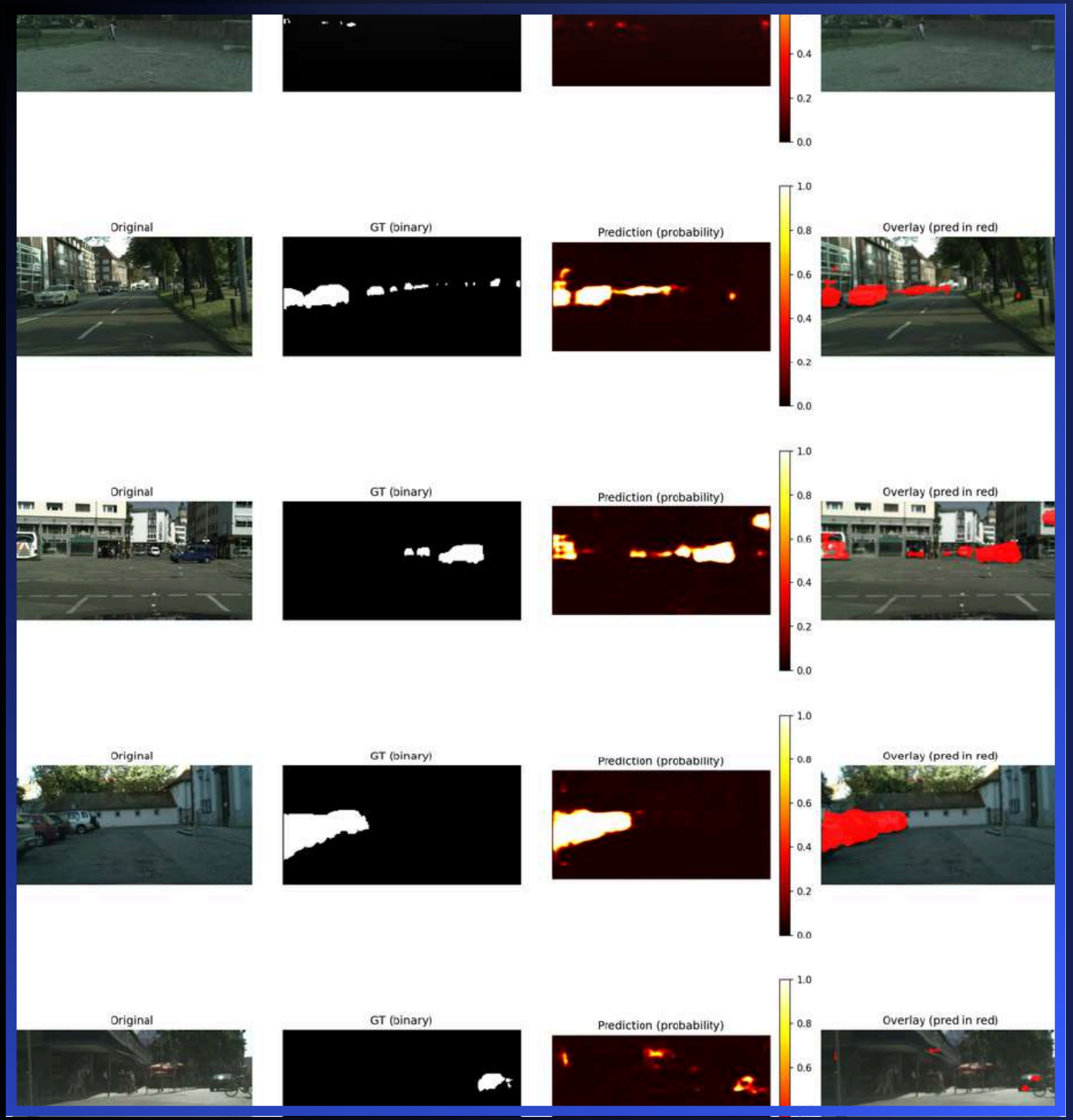
COMPARISON

Compare Based on AUC (Area Under Curve)
→ Measures the model's ability to discriminate between classes (person vs. background)

Heavier model has better results
→ complex architecture needed to overcome limitations

SOME EXTRAS


Original
GT (binary)
Prediction (probability)
Overlay (pred in red)

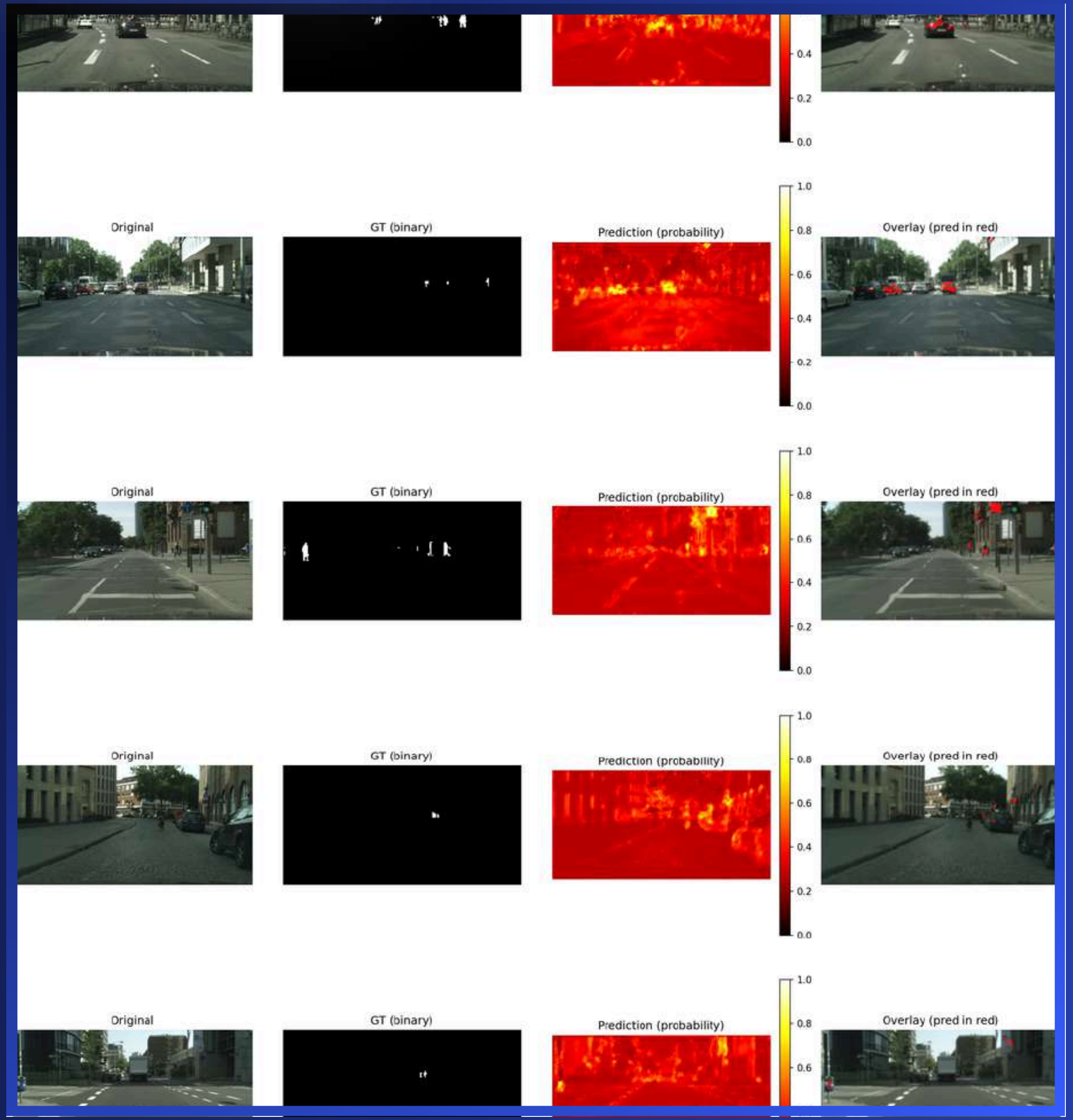

Original
GT (binary)
Prediction (probability)
Overlay (pred in red)


Image 76
Ground truth
Prediction
Image 31
Ground truth
Prediction
Image 93
Ground truth
Prediction
Image 23
Ground truth
Prediction


Original Image
Binary Mask
(White = Person Detected)
People Coloured Out

THANK YOU